IGCSE 0607 Practice Questions — E8.1 — Combined Questions & Markscheme

Transformations

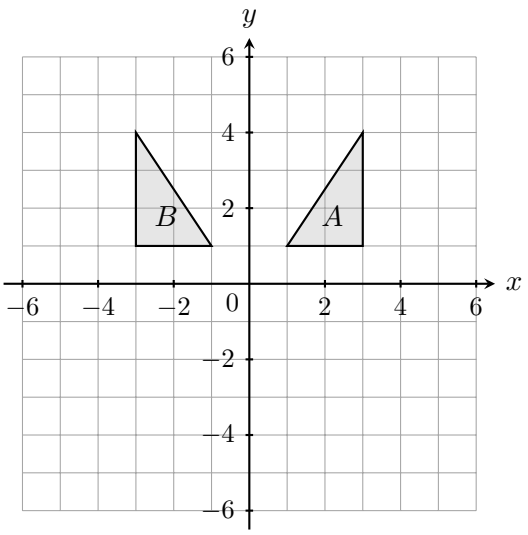
Total Questions: 15 — **Total Marks:** 83 Marks

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- **E8.1** Transformations
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

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1 The grid shows triangle A and triangle B.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -2 \\ -5 \end{pmatrix}$. **[2]**

(b) Describe fully the single transformation that maps triangle A onto triangle B.

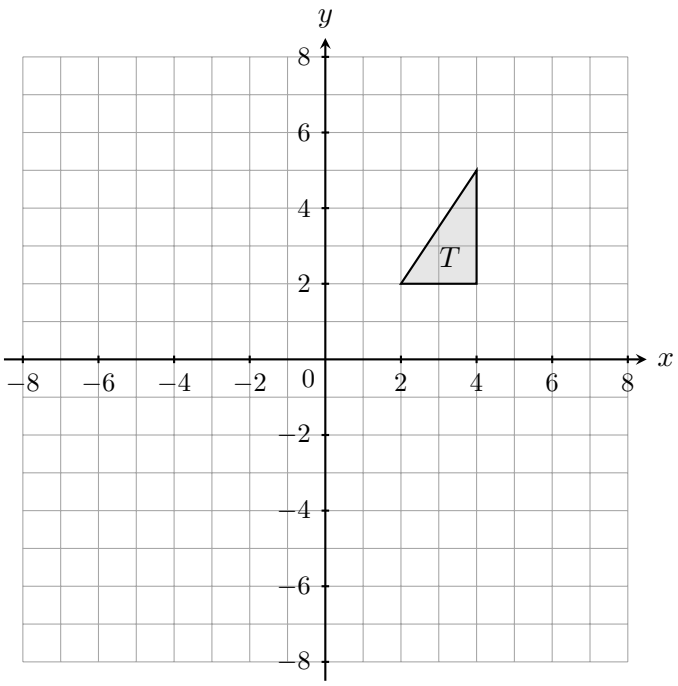
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[2]

2 The grid shows triangle T.



(a) On the grid, draw the image of triangle T after an enlargement with scale factor $-\frac{1}{2}$ and centre $(0, 2)$. **[2]**

(b) Triangle T is reflected in the line $x = 1$ to give triangle U.
Triangle U is then reflected in the line $y = 2$ to give triangle V.
Describe fully the single transformation that maps triangle T onto triangle V.

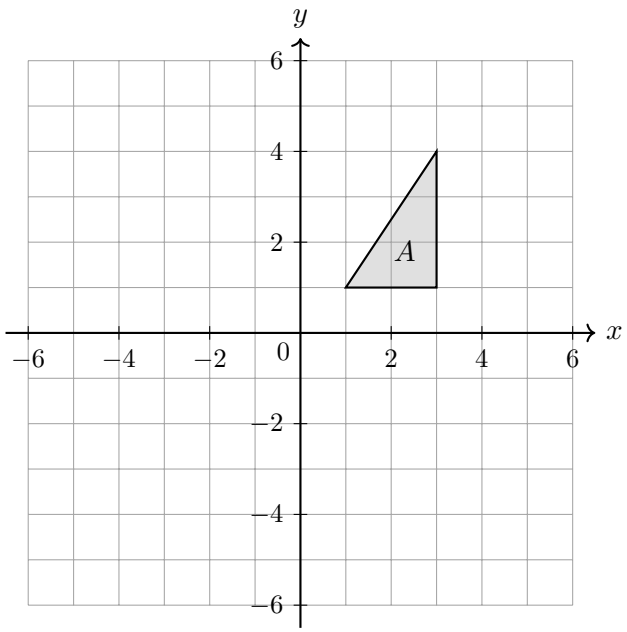
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[3]

(c) Triangle T is enlarged by a scale factor of k to give triangle W.
The area of triangle W is 48 square units.
Find the value of k, given that $k > 0$.

$k =$ ..... **[2]**

3 The diagram shows triangle A on a coordinate grid.

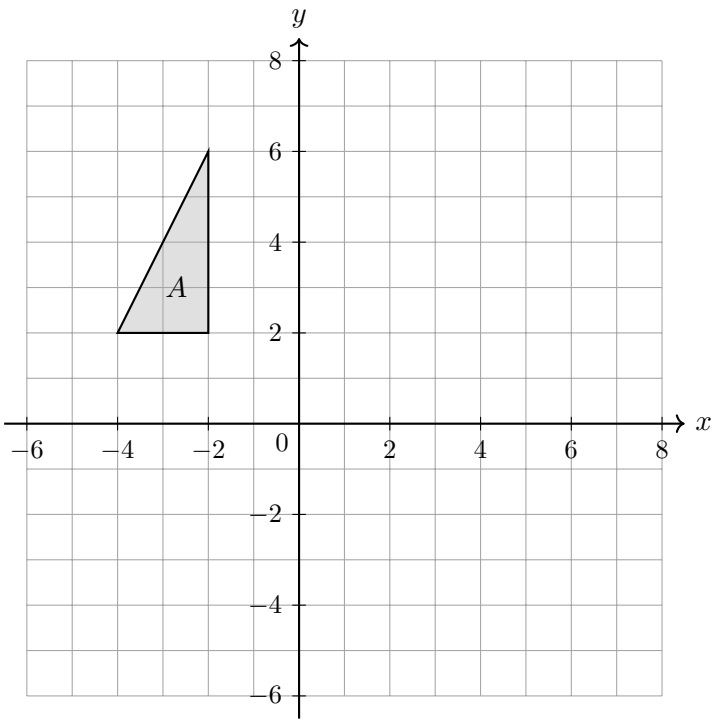


(a) On the grid, translate shape A by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$.
Label the image B. **[2]**

(b) On the grid, reflect shape A in the x-axis.
Label the image C. **[2]**

(c) On the grid, rotate shape A by 180° about the origin $(0, 0)$.
Label the image D. **[2]**

4 The diagram shows shape A on a coordinate grid.



(a) Shape A is reflected in the line $y = x$ to give shape B.
On the grid, draw and label shape B. **[2]**

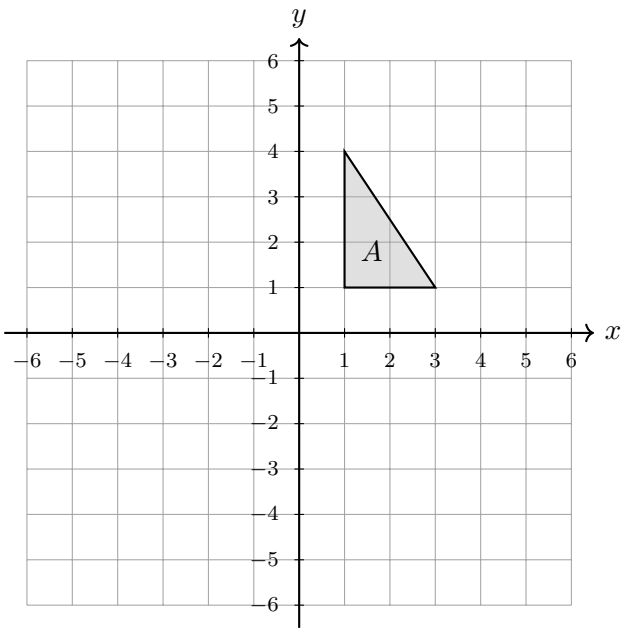
(b) Shape B is rotated by 90° clockwise about the point $(2, 2)$ to give shape C.
On the grid, draw and label shape C. **[2]**

(c) Describe fully the single transformation that maps shape A onto shape C.

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..... **[2]**

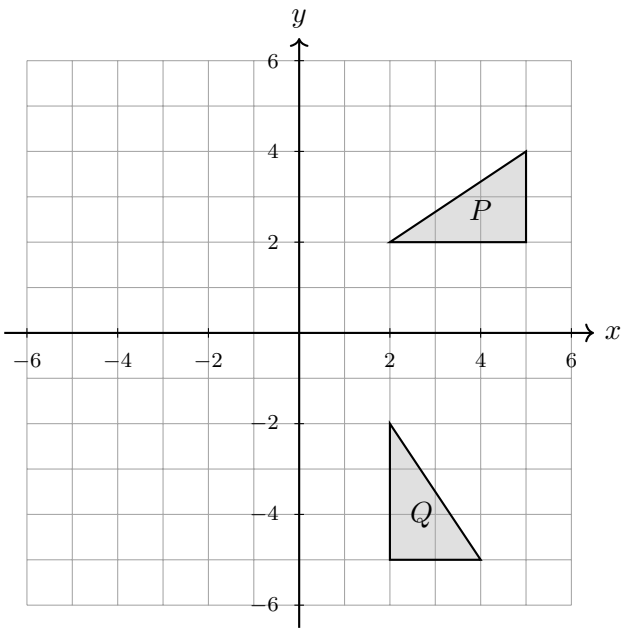
5 The diagram shows a triangle, Shape A, drawn on a coordinate grid.



(a) On the grid, draw the image of Shape A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label this image B. **[2]**

(b) On the grid, draw the image of Shape A after a reflection in the line $y = -1$.
Label this image C. **[2]**

6 The diagram shows two triangles, Shape P and Shape Q, drawn on a grid.



(a) Describe fully the single transformation that maps Shape P onto Shape Q.

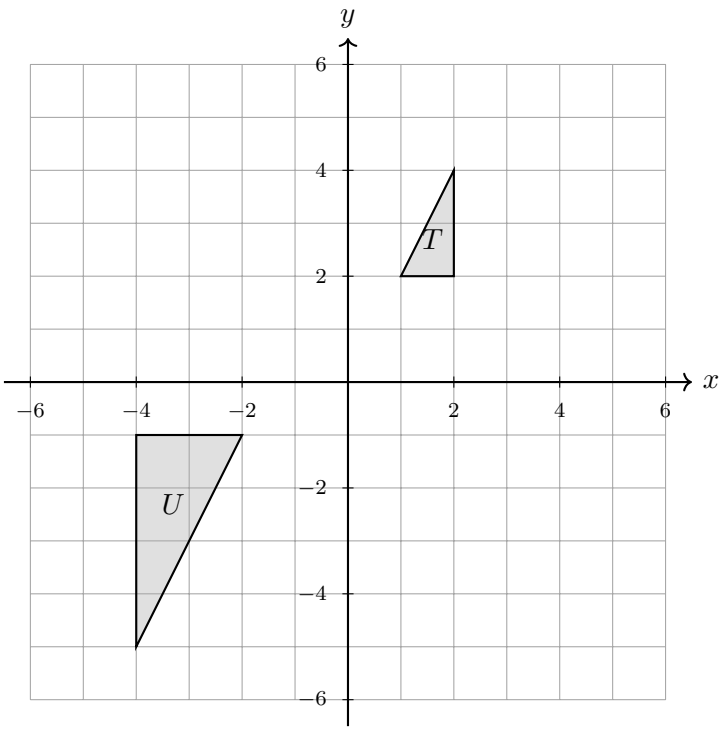
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..... **[3]**

(b) Shape P is enlarged by a scale factor of $\frac{1}{2}$ with centre of enlargement $(-2, 2)$.
On the grid, draw the image of Shape P and label it R. **[2]**

7 The diagram shows two triangles, Shape T and Shape U.



(a) Describe fully the single transformation that maps Shape T onto Shape U.

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..... **[3]**

(b) Describe fully the single transformation that maps Shape U onto Shape T.

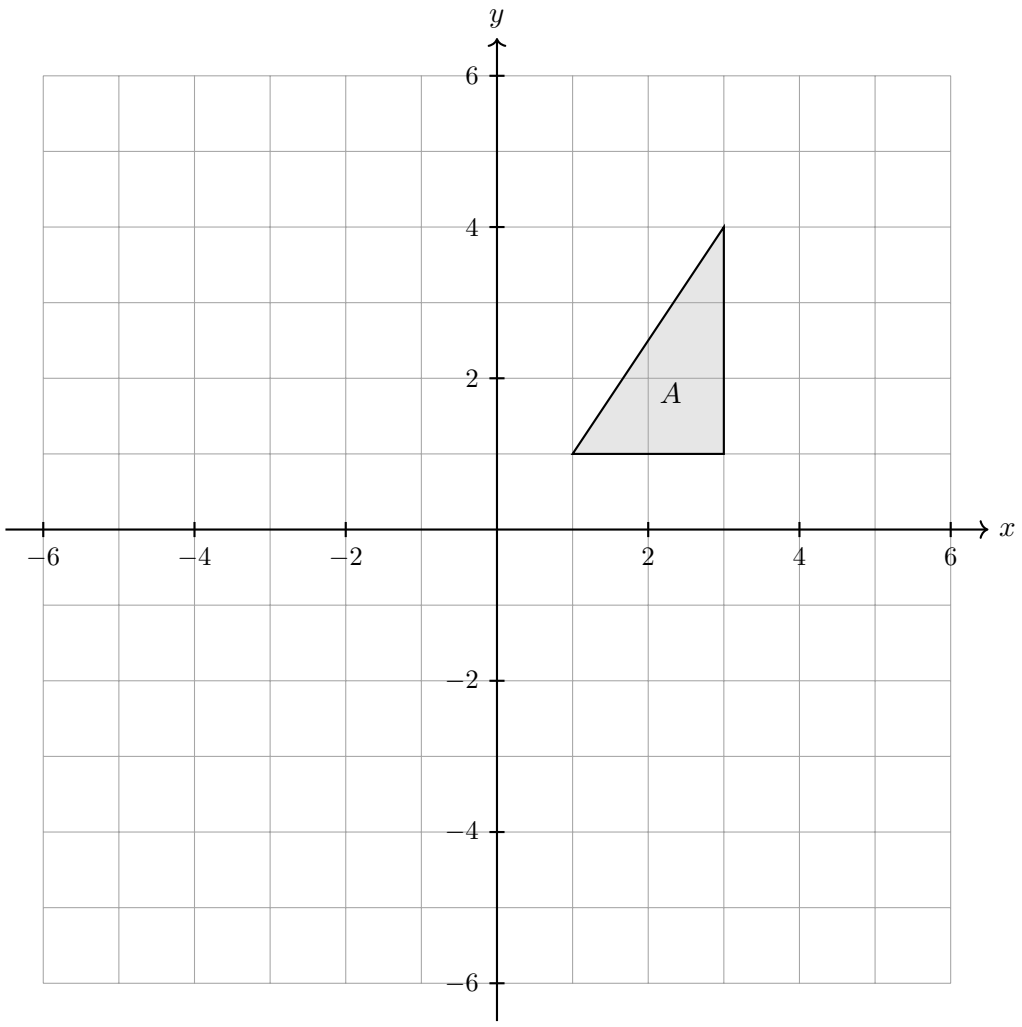
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..... **[2]**

(c) Shape T is translated by the vector $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$ to give Shape V.
Shape V is then reflected in the line $y = x$ to give Shape W.
Find the coordinates of the three vertices of Shape W.

..... **[3]**

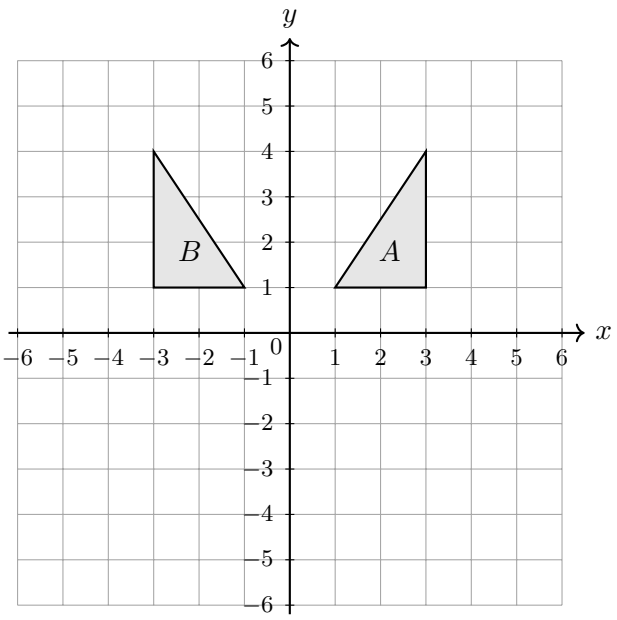
8 The grid shows triangle A.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label the image B. **[2]**

(b) On the grid, draw the image of triangle A after a reflection in the line $x = -1$.
Label the image C. **[2]**

9 The diagram shows two triangles, A and B, plotted on a coordinate grid.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$. **[2]**

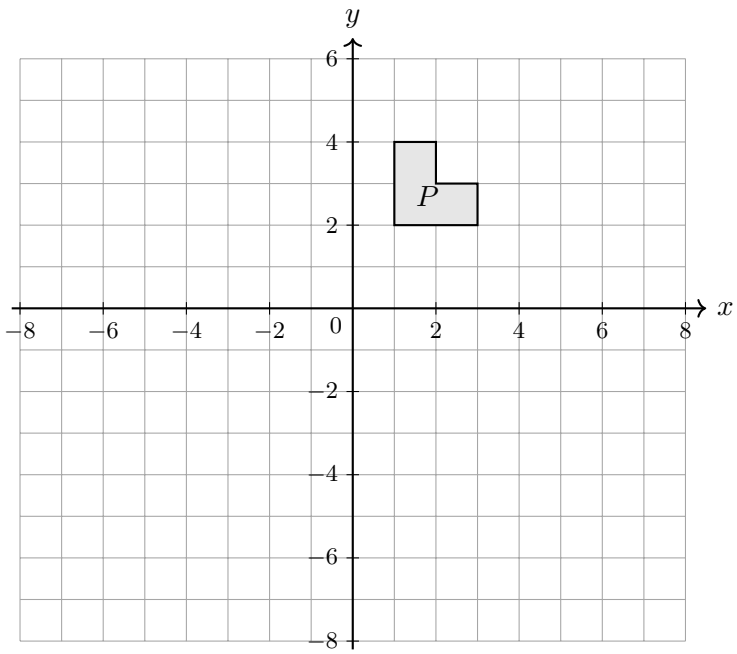
(b) Describe fully the single transformation that maps triangle A onto triangle B.

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..... **[2]**

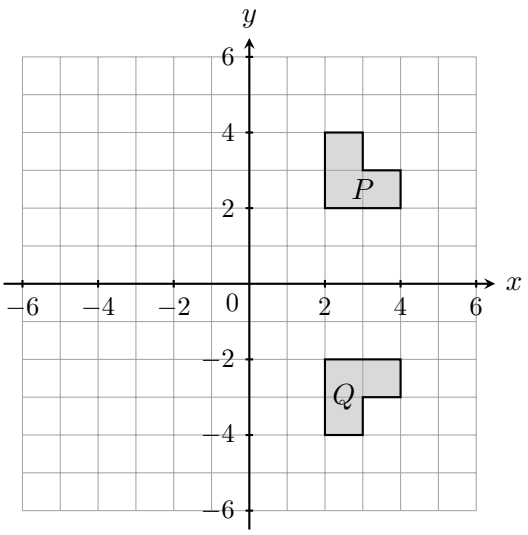
10 The diagram shows a shape P plotted on a coordinate grid.



(a) On the grid, draw the image of shape P after an enlargement, scale factor -2, centre $(0, 0)$. **[2]**

(b) On the grid, draw the image of shape P after a rotation of $90°$ clockwise about the point $(1, -1)$. **[2]**

11 The grid shows shapes P and Q.



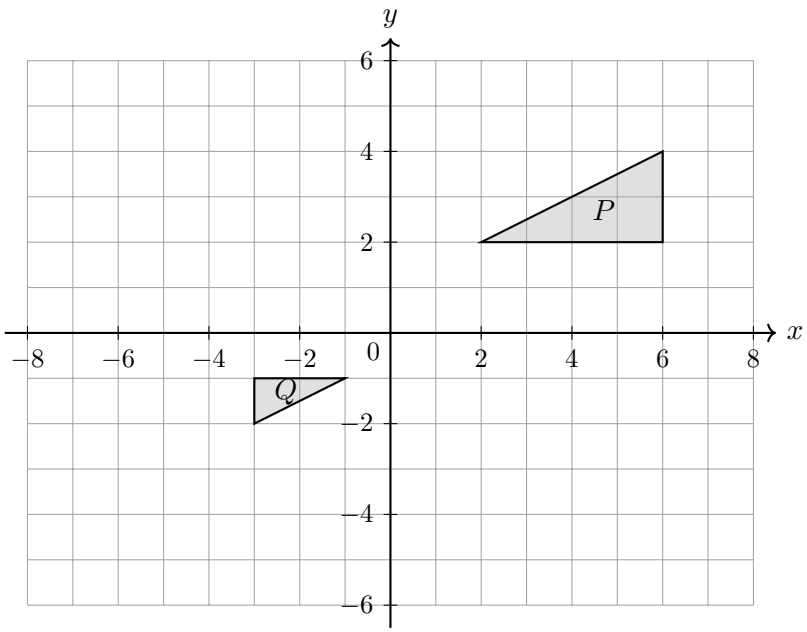
(a) On the grid, draw the image of shape P after an enlargement with scale factor 0.5 and centre $(0, 0)$. **[2]**

(b) Describe fully the single transformation that maps shape P onto shape Q.

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[3]

12 The diagram shows triangles P and Q on a coordinate grid.



(a) Describe fully the single transformation that maps shape P onto shape Q.

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..... **[3]**

(b) On the grid, enlarge shape P by scale factor $\frac{1}{2}$ with centre $(0, 4)$.
Label the image R. **[3]**

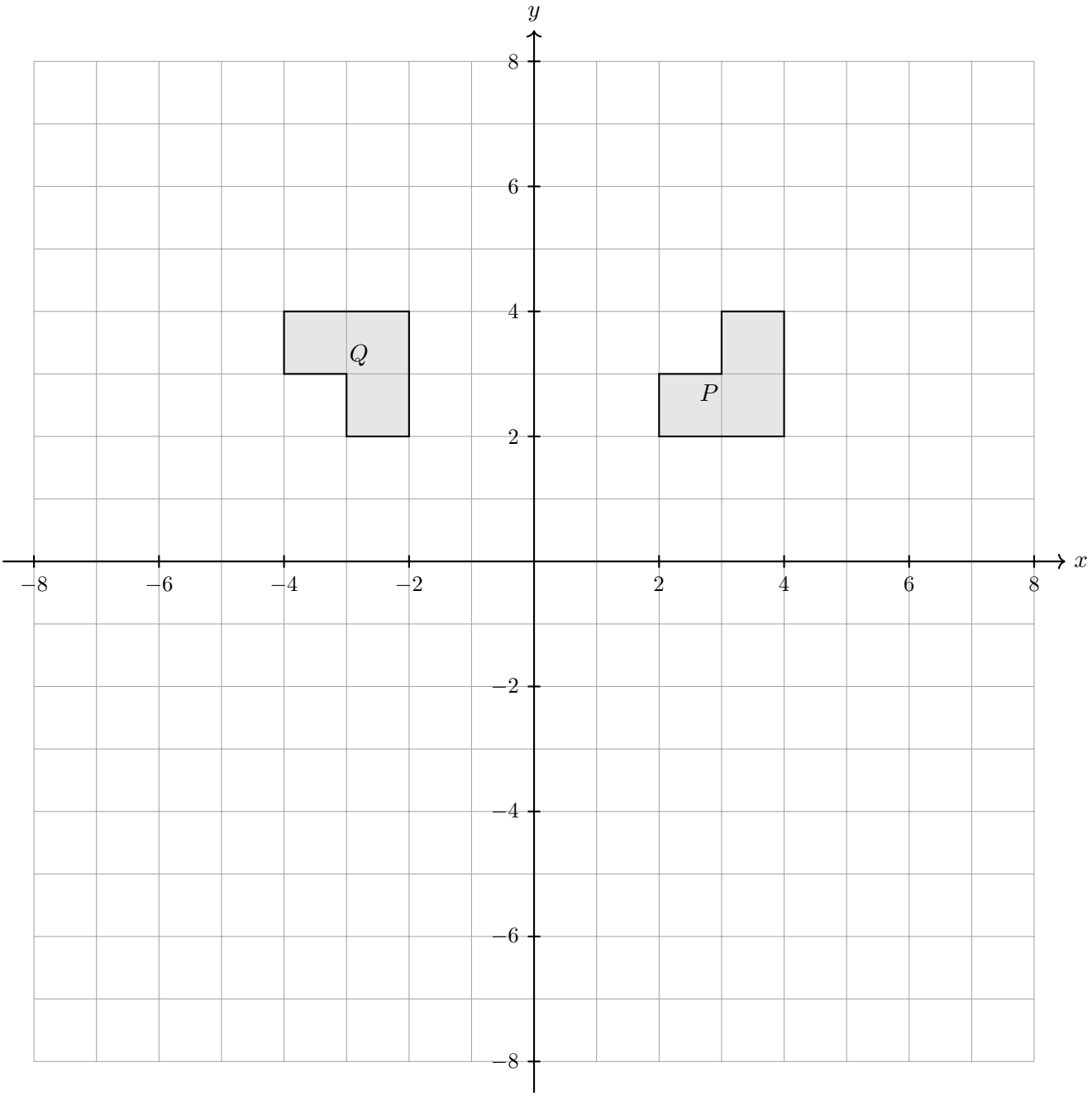
(c) Describe fully the single transformation that maps shape R onto shape Q.

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..... **[3]**

13 The grid shows shapes P and Q.



(a) Describe fully the single transformation that maps shape P onto shape Q.

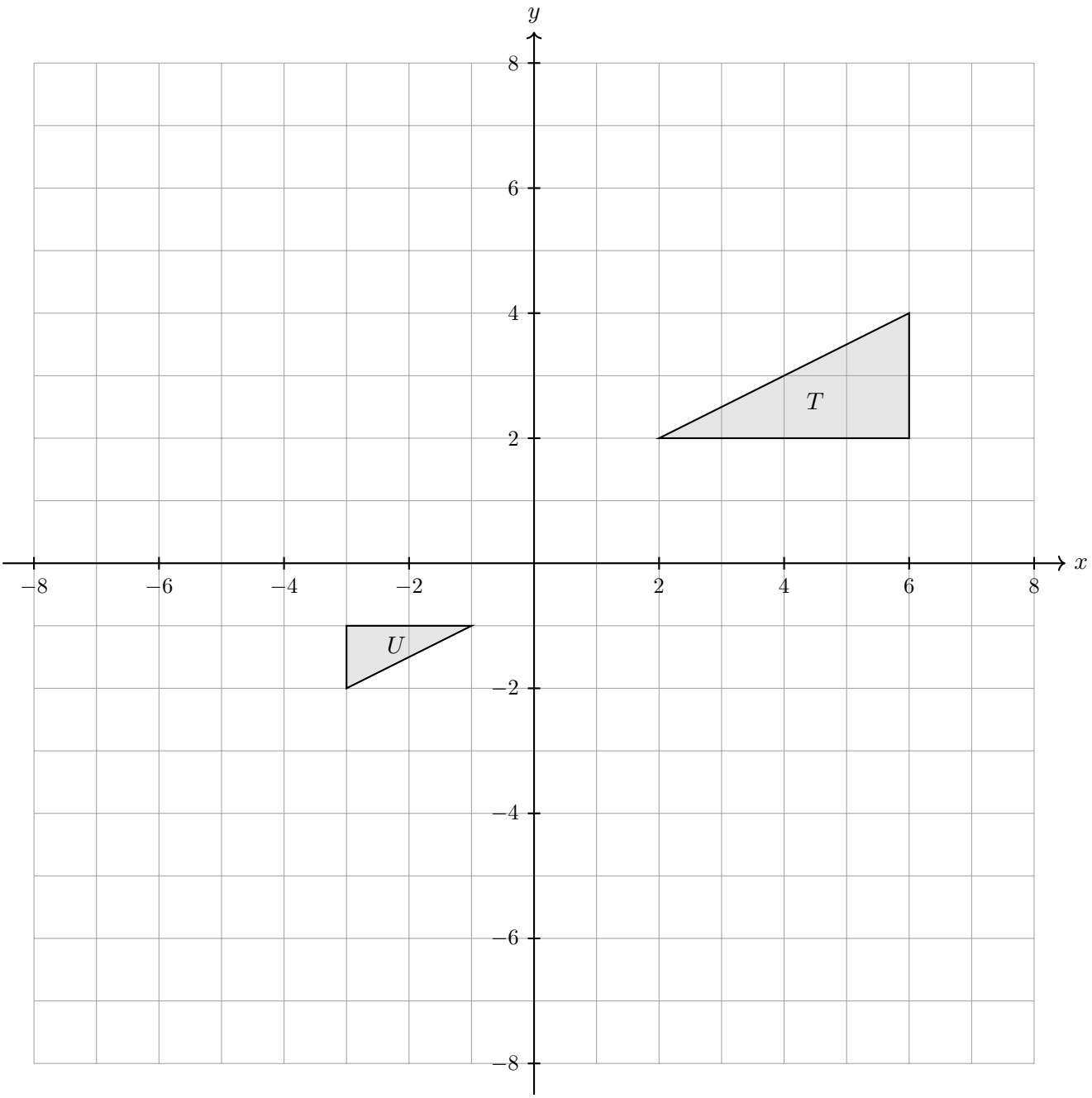
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..... **[3]**

(b) On the grid, enlarge shape P by scale factor $\frac{1}{2}$, centre $(6, 6)$.
Label the image R. **[2]**

14 The grid shows triangles T and U.



(a) Describe fully the single transformation that maps triangle T onto triangle U.

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..... **[3]**

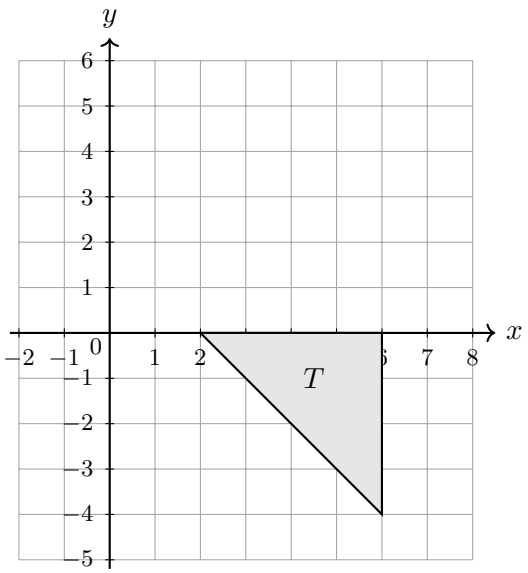
(b) Triangle T is reflected in the line $y = x$ to give triangle V.
Triangle V is then rotated 90° clockwise about the origin to give triangle W.
Describe fully the single transformation that maps triangle T onto triangle W.

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..... **[2]**

15 The diagram shows triangle T plotted on a coordinate grid.



(a) Triangle T is mapped onto triangle U by an enlargement, scale factor $\frac{1}{2}$, centre $(0, 4)$.
On the grid, draw triangle U. **[2]**

(b) Triangle U is mapped onto triangle V by a reflection in the line $y = x$.
On the grid, draw triangle V. **[2]**

(c) Describe fully the single transformation that maps triangle U back onto triangle T.

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..... **[3]**

1 **(a)** Vertices of image plotted correctly at $(-1,-4)$, $(1,-4)$ and $(1,-1)$ **[B2]**

If not B2, award B1 for correct size and orientation, or 2 vertices correct **[B1]**

(b) Reflection **[B1]**

in the y-axis (or $x=0$) **[B1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Vertices of image plotted correctly at $(-1,2)$, $(-2,2)$ and $(-2,0.5)$ **[B2]**

If not B2, award B1 for enlargement SF $-\frac{1}{2}$ from wrong centre, or correct centre but SF $\frac{1}{2}$ **[B1]**

(b) Rotation **[B1]**

$180°$ (direction not required) **[B1]**

centre $(1,2)$ **[B1]**

(c) Calculates area of $T = \frac{1}{2} \times 2 \times 3 = 3$ **[M1]**

Sets up $3 \times k^2 = 48 \implies k^2 = 16$ **[M1]**

$k = 4$ **[A1]**

[Back to Question] *[Total Marks: 7]*

3 **(a)** Image B drawn correctly at $(-4,3), (-2,3), (-2,6)$ **[B2]**

If incorrect, 1 mark for translation by $\begin{pmatrix} -5 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$ **[M1]**

(b) Image C drawn correctly at $(1,-1), (3,-1), (3,-4)$ **[B2]**

If incorrect, 1 mark for reflection in y-axis or in a line $y = k$ **[M1]**

(c) Image D drawn correctly at $(-1,-1), (-3,-1), (-3,-4)$ **[B2]**

If incorrect, 1 mark for rotation of $180°$ about the wrong centre, or $90°$ about $(0,0)$ **[M1]**

[Back to Question] *[Total Marks: 6]*

4 **(a)** Image B drawn correctly at $(2,-4), (2,-2), (6,-2)$ **[B2]**

If incorrect, 1 mark for reflection in $y = -x$ or correct orientation but wrong position **[M1]**

(b) Image C drawn correctly at $(-4,2), (-2,2), (-2,-2)$ **[B2]**

If incorrect, 1 mark for rotation $90°$ anti-clockwise about $(2,2)$ or $90°$ clockwise about the wrong centre **[M1]**

(c) Reflection **[B1]**

in the line $y = 2$ **[B1]**

[Back to Question] *[Total Marks: 6]*

5 **(a)** Correct translation of Shape A **[B2]**

If incorrect, translation of $\begin{pmatrix} -4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$ where $k \neq 0$ **[B1]**

(b) Correct reflection of Shape A to form Shape C with vertices at $(1, -3)$, $(3, -3)$, and $(1, -6)$ **[B2]**

If incorrect, reflection in $x = -1$ or $y = 1$ **[B1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** Rotation **[B1]**

$90°$ clockwise OR $270°$ anti-clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Correct triangle drawn with vertices at $(0, 2)$, $(1.5, 2)$, and $(1.5, 3)$ **[B2]**

If incorrect, correct size and orientation but wrong position OR correct centre of enlargement but scale factor 2 **[B1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(0, 1)$ **[B1]**

(b) Enlargement scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0, 1)$ **[B1]**

(c) Translating Shape T gives at least two correct coordinates of V: $(-2, 3)$, $(-1, 3)$, $(-1, 5)$ **[M1]**

Recognising reflection in $y = x$ swaps coordinates (x, y)

(y, x)$[M1]$

Final coordinates: $(3, -2)$, $(3, -1)$, $(5, -1)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

8 **(a)** Correct triangle B with vertices at $(-3, 3), (-1, 3), (-1, 6)$ **[B2]**

Translation by vector $\begin{pmatrix} -4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$ **[B1]**

(b) Correct triangle C with vertices at $(-3, 1), (-5, 1), (-5, 4)$ **[B2]**

Reflection in $y = -1$ or correct orientation but wrong position **[B1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** Correct triangle with vertices at $(-3, 3), (-1, 3)$ and $(-1, 6)$ **[B2]**

Translation by correct vector but incorrect shape/orientation OR correct size, shape and orientation but wrong translation **[B1]**

(b) Reflection **[B1]**

in the y-axis OR $x = 0$ **[B1]**

[Back to Question] *[Total Marks: 4]*

10 (a) Correct shape with vertices at $(-2, -4), (-6, -4), (-6, -6), (-4, -6), (-4, -8)$ and $(-2, -8)$ **[B2]**

Enlargement, scale factor 2 centre $(0, 0)$ OR correct scale factor and orientation but wrong centre **[B1]**

(b) Correct shape with vertices at $(4, -1), (4, -3), (5, -3), (5, -2), (6, -2)$ and $(6, -1)$ **[B2]**

Rotation of 90° clockwise about the wrong centre OR rotation of 90° anti-clockwise about $(1, -1)$ **[B1]**

[Back to Question] *[Total Marks: 4]*

11 (a) Vertices of image plotted correctly at $(1, 1)$, $(2, 1)$, $(2, 1.5)$, $(1.5, 1.5)$, $(1.5, 2)$ and $(1, 2)$ **[B2]**

If not B2, award B1 for correct size and orientation, or correct enlargement from the wrong centre **[B1]**

(b) Rotation **[B1]**

90° clockwise (or −90°) **[B1]**

centre $(0, 0)$ (or origin) **[B1]**

[Back to Question] *[Total Marks: 5]*

12 (a) Enlargement **[B1]**

Centre $(0, 0)$ or origin **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

(b) Image R drawn correctly at $(1, 3), (3, 3), (3, 4)$ **[B3]**

If incorrect, 2 marks for correct size and orientation but wrong position **[M2]**

1 mark for correct centre used but wrong scale factor **[M1]**

(c) Rotation **[B1]**

180° **[B1]**

Centre $(0, 1)$ **[B1]**

OR Enlargement [B1], Scale factor −1 [B1], Centre $(0, 1)$ **[B1]**

[Back to Question] *[Total Marks: 9]*

13 (a) Rotation **[B1]**

$90°$ anticlockwise OR $270°$ clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Correct shape R with vertices at $(4, 4), (5, 4), (5, 5), (4.5, 5), (4.5, 4.5), (4, 4.5)$ **[B2]**

Any enlargement scale factor $\frac{1}{2}$ OR correct centre $(6, 6)$ with incorrect scale factor **[B1]**

[Back to Question] *[Total Marks: 5]*

14 (a) Enlargement **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Reflection **[B1]**

in the x-axis OR $y = 0$ **[B1]**

[Back to Question] *[Total Marks: 5]*

15 (a) Correct triangle U with vertices at $(1, 2), (3, 2)$ and $(3, 0)$ **[B2]**

Enlargement scale factor $\frac{1}{2}$ from wrong centre OR ray lines drawn from $(0, 4)$ but scale factor incorrect **[B1]**

(b) Correct triangle V with vertices at $(2, 1), (2, 3)$ and $(0, 3)$ **[B2]**

Reflection of U in the line $y = -x$ OR reflection of T in the line $y = x$ **[B1]**

(c) Enlargement **[B1]**

Scale factor 2 **[B1]**

Centre $(0, 4)$ **[B1]**

[Back to Question] *[Total Marks: 7]*